

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Agroecosystems: abiotic & biotic factors - Set 1

Mark scheme

AQA spec 3.5.1.1; Biological resources; textbook pp. 306-320

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• B: frost-free period (150 days) exceeds the 120 needed whereas A (110) is too short - late frosts would kill seedlings • B is warmer / longer thermal growing season because south-facing slopes receive more insolation; valley floors are frost pockets where cold dense air collects • But B has lower rainfall and a steeper slope (erosion / machinery difficulties) so irrigation / contour cultivation may be needed	3
01.2	<i>Any 3 from:</i> • Greenhouses / polytunnels heated by solar gain or burning gas / paraffin (ventilated to prevent overheating) • Woven plastic cloches / fleece over rows to retain warm air near the ground • Burning oil / gas or using large fans in orchards to prevent frost damage to blossom • Site selection (aspect, avoiding frost pockets); heated buildings and shelter for livestock	3
01.3	• Mammals and birds use food energy to maintain a constant body temperature (thermoregulation) • Reducing heat loss (shelter, heated buildings) means less food energy is used for heat and more is available for growth	2
01.4	• Long days increase milk production and egg laying; poultry grow best with short days; sheep mate as days shorten in autumn • Artificial lighting extends the growing season for greenhouse crops or simulates autumn in spring to give sheep a second mating / lambing season	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-309; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
02.1	- Plants close their stomata to reduce transpiration losses - Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop - Less water for nutrient absorption / transport and cell turgidity; severe shortage dehydrates cells and inhibits biochemical reactions	3
02.2	- Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms; higher risk of fungal disease; root respiration inhibited - Drainage ditches or pipes - Deep ploughing; avoiding compaction by machinery / livestock; adding organic matter to encourage worms	4
02.3	- Groundwater often contains dissolved salts; when the water evaporates from the soil the salts are left behind and accumulate - High soil salinity causes osmotic dehydration of roots / crop death; washing out with extra water uses more water and salinises rivers	2
02.4	Add crushed lime (calcium carbonate / hydroxide); low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	
03.1	- Organic advantages: often waste products, locally available and cheap; add humus improving structure and water retention; increase soil biota; slow release reduces leaching - Organic disadvantages: nutrient composition cannot be controlled; released slowly so needs long-term planning; bulky with high water content so expensive to transport; cannot be applied to a growing crop; odour / pathogens - Inorganic advantages: composition controlled to match crop needs; nutrients released rapidly; concentrated and easy to apply - Inorganic disadvantages: high energy (fossil fuel) manufacture (Haber process); no organic matter so humus and biota decline; some toxic to worms; high solubility so leached causing eutrophication; raw materials (phosphate rock) finite	6
03.2	- Crops grown in a circulating nutrient solution (inert medium) in greenhouses; nutrient concentrations monitored and adjusted so nutrients are never limiting; abiotic factors controlled - Roots are small (all in contact with nutrients) so more growth goes into the harvested crop; no soil-borne pathogens or weeds; crop stays fresh with roots - Disadvantages: high inputs of nutrients and energy (heating, lighting, pumping); high level of technical knowledge; only economic for high-value crops	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	
04.1	- Rotation: each crop has its own pests; growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early - Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids - Rotation also evens out nutrient demands and includes legumes	4

Question	Marking guidance	Marks
04.2	- Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) - It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 - Cultural control alone loses about 12% yield; integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact	3
04.3	- Large numbers of males are sterilised with gamma radiation and released - Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring; enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) - Only the target species' females mate with them so non-target species are unaffected; fails if sterilised males behave abnormally or re-colonisation occurs	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	
05.1	- Success: Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia; Encarsia parasitic wasps control greenhouse whitefly - Problems: cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators; rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct); harlequin ladybirds introduced to Europe eat native ladybirds - Lesson: control agents should be specialist feeders on the pest	4
05.2	- Release an artificial sex scent attracting males (or females) of the pest species - Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding	2
05.3	- Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply - Agricultural use exceeds medical use, so resistance spreads - Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat; hence reduced use of antibiotics is a sustainability goal	3
05.4	Grow nectar-rich flowering plants alongside crops; restrict pesticides harmful to bees / spray when not in flower; introduce bee hives	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.	